Coefficientwise log-concavity of the normalized incomplete beta function

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Task 1. Resolve the first-parameter conjecture

For $a, b > 0$, let $I_x(a, b)$ be the regularized incomplete beta function and set

$$f_b(a; x) = x^{-a}(1-x)^{-b} I_x(a, b) = \frac{1}{\Gamma(b)} \sum_{n=0}^{\infty} \frac{\Gamma(a+b+n)}{\Gamma(a+1+n)} x^n, \qquad |x| < 1.$$

The series identity follows from Euler's transformation of the standard hypergeometric representation of I_x; it also identifies the formal coefficients under consideration. Write

$$f_b(a+\alpha; x) f_b(a+\beta; x) - f_b(a; x) f_b(a+\alpha+\beta; x) = \sum_{m=0}^{\infty} \psi_m x^m.$$

The following proves Conjecture 2 of Karp [1], including the unrestricted positive shifts in the previously unsettled $b > 1$ case.

Theorem 1. *For every $a, \alpha, \beta > 0$ and every integer $m \geq 0$,*

$$\psi_m > 0 \quad (b > 1), \qquad \psi_m < 0 \quad (0 < b < 1), \qquad \psi_m = 0 \quad (b = 1).$$

Thus $a \mapsto f_b(a; x)$ is strictly coefficientwise Wright log-concave for $b > 1$ and strictly coefficientwise Wright log-convex for $0 < b < 1$.

The $0 < b < 1$ assertion was already proved in the source's Corollary 1. We give its short moment proof below to make the result self-contained. The main step for $b > 1$ is a finite-sum inequality.

A finite-sum inequality

Put

$$g_b(t) = \frac{\Gamma(t+b)}{\Gamma(t+1)}, \qquad t > 0.$$

Lemma 1. *Let $b > 1$, $u > v > 0$ and $m \geq 0$ be an integer. Then*

$$\sum_{k=0}^{m} g_b(u+k) g_b(v+m-k) \left(q^{u+k} - q^{v+m-k}\right) < 0 \qquad (0 < q < 1). \tag{1}$$

Proof. Set $T = u + v + m$, $w(t) = g_b(t)g_b(T-t)$ and

$$P_u(q) = \sum_{k=0}^{m} w(u+k)q^{u+k}, \qquad P_v(q) = \sum_{k=0}^{m} w(v+k)q^{v+k}, \qquad F = P_u - P_v.$$

Since $w(t) = w(T-t)$, F is the left side of (1). Let $\theta = q\,d/dq$ and introduce the differential operator

$$\mathcal{L} = \theta(T+b-\theta) - q(\theta+b)(T-\theta).$$

The weight recurrence

$$(t+1)(T-t+b-1)w(t+1) = (t+b)(T-t)w(t) \tag{2}$$

shows that all interior terms in $\mathcal{L}P_u$ cancel. Subtracting the two boundary expressions gives

$$\mathcal{L}F = R(q) := C\big(q^u + q^{T-u+1}\big) - D\big(q^v + q^{T-v+1}\big), \tag{3}$$

where

$$C = u(T+b-u)w(u), \qquad D = v(T+b-v)w(v).$$

We first determine the sign changes of R. Define $r_b(t) = \Gamma(t+b)/\Gamma(t)$. Its logarithm is strictly concave, since

$$(\log r_b)''(t) = \psi'(t+b) - \psi'(t) < 0;$$

the last inequality follows, for example, from $\psi'(t) = \sum_{j\ge 0}(t+j)^{-2}$. Moreover,

$$C = r_b(u)r_b(T+1-u), \qquad D = r_b(v)r_b(T+1-v).$$

The two arguments in either product have sum $T+1$, and

$$\left|u - \frac{T+1}{2}\right| < \left|v - \frac{T+1}{2}\right|,$$

because $u > v$ and $u + v = T - m < T + 1$. Strict log-concavity therefore gives $C > D > 0$.

The ratio

$$\frac{q^u + q^{T-u+1}}{q^v + q^{T-v+1}} = \frac{\cosh(\rho s)}{\cosh(\sigma s)}, \qquad s = -\log q, \quad \rho = \left|u - \frac{T+1}{2}\right|, \quad \sigma = \left|v - \frac{T+1}{2}\right|,$$

increases strictly from 0 to 1 as q increases from 0 to 1. Indeed $0 \le \rho < \sigma$, and its logarithmic derivative in $s > 0$ is $\rho\tanh(\rho s) - \sigma\tanh(\sigma s) < 0$. Thus R has exactly one zero in $(0,1)$, with signs $-, +$.

We next construct a positive solution of the homogeneous equation. Let

$$Y(q) = q^{T+b}h(q), \qquad h(q) = \sum_{j=0}^{\infty} \frac{(b)_j (T+2b)_j}{(T+b+1)_j\, j!} q^j. \tag{4}$$

The coefficients are positive and the series converges for $0 < q < 1$. Its coefficient recurrence gives $\mathcal{L}Y = 0$ directly. In particular $Y(q) > 0$ and $Y(q) \ge q^{T+b}$.

For use at the endpoints, the beta integral gives

$$h(q) = \frac{1}{B(b, T+1)} \int_0^1 \frac{t^{b-1}(1-t)^T}{(1-qt)^{T+2b}}\, dt. \tag{5}$$

Near zero, $Y = O(q^{T+b})$ and $Y' = O(q^{T+b-1})$. Near one, writing $\varepsilon = 1 - q$ and using $q \geq 1/2$, $t^{b-1} \leq 1$ and $1 - qt \geq (\varepsilon + 1 - t)/2$, we obtain

$$h(q) = O(\varepsilon^{1-2b}), \qquad h'(q) = O(\varepsilon^{-2b}).$$

For example, the first estimate follows by bounding the integral by a constant times $\int_0^1 z^T(\varepsilon + z)^{-T-2b}\,dz$ and substituting $z = \varepsilon y$; the resulting integral on $(0, \infty)$ is finite because $b > 1$. Differentiating (5) proves the second estimate in the same way. Consequently

$$Y = O((1-q)^{1-2b}), \qquad Y' = O((1-q)^{-2b}) \quad (q \uparrow 1). \tag{6}$$

In ordinary derivatives, equation (3) is

$$-q^2(1-q)F'' + q\big(T + b - 1 - q(T - b - 1)\big)F' - qbTF = R.$$

Set

$$p(q) = q^{1-T-b}(1-q)^{2b}, \qquad W(q) = p(q)\big(Y(q)F'(q) - Y'(q)F(q)\big).$$

The homogeneous equation for Y and the displayed equation for F give the exact identity

$$W'(q) = -\frac{p(q)Y(q)}{q^2(1-q)}R(q). \tag{7}$$

Both endpoint limits of W are zero. At zero, $F = O(q^v)$, $F' = O(q^{v-1})$, so $W = O(q^v)$. At one, the finite sums give $F(1) = 0$, $F = O(1-q)$ and $F' = O(1)$; using (6) gives $W = O(1-q)$.

Since R has signs $-, +$ and all other factors in (7) are positive, W first increases from zero and then decreases to zero. Hence $W(q) > 0$ throughout $(0, 1)$. It follows that

$$\left(\frac{F}{Y}\right)' = \frac{W}{pY^2} > 0.$$

Finally $F/Y \to 0$ as $q \uparrow 1$, since $F \to 0$ and $Y \geq q^{T+b}$. Thus $F/Y < 0$, and $F < 0$, as required. □

Proof of coefficientwise log-concavity

For fixed m define the symmetric function

$$H_m(u, v) = \sum_{k=0}^{m} g_b(u + k)g_b(v + m - k).$$

Suppose $b > 1$ and hold $u + v$ fixed. Differentiating with respect to u (so $dv/du = -1$) and using the digamma integral gives

$$\frac{d}{du}H_m(u, v) = \int_0^\infty \frac{e^{-s} - e^{-bs}}{1 - e^{-s}} \sum_{k=0}^{m} g_b(u + k)g_b(v + m - k)\big(e^{-(u+k)s} - e^{-(v+m-k)s}\big)\,ds. \tag{8}$$

Indeed

$$\frac{d}{dt}\log g_b(t) = \psi(t + b) - \psi(t + 1) = \int_0^\infty e^{-ts}\frac{e^{-s} - e^{-bs}}{1 - e^{-s}}\,ds.$$

There are only finitely many terms, and the displayed integrals converge absolutely. For $u > v$, the finite sum in (8) is strictly negative by Lemma 1, with $q = e^{-s}$. The remaining factor is strictly positive. Thus $H_m(u, v)$ strictly decreases as $|u - v|$ increases at fixed $u + v$.

The pairs $(a+\alpha, a+\beta)$ and $(a, a+\alpha+\beta)$ have the same sum, whereas $|\alpha-\beta| < \alpha+\beta$. Therefore

$$\psi_m = \frac{H_m(a + \alpha, a + \beta) - H_m(a, a + \alpha + \beta)}{\Gamma(b)^2} > 0 \qquad (b > 1).$$

This applies also to $m = 0$ and requires no integrality of either shift.

The other values of b

For $0 < b < 1$, the beta integral gives a positive measure

$$d\nu_b(s) = \frac{s^{b-1}(1-s)^{-b}}{\Gamma(1-b)}\,ds, \qquad 0 < s < 1, \qquad g_b(t) = \int_0^1 s^t\,d\nu_b(s).$$

Put $K_m(s,t) = \sum_{k=0}^m s^k t^{m-k}$, which is positive and symmetric on $(0,1)^2$. Multiplying the moment integrals and symmetrizing yields

$$\psi_m = -\frac{1}{2\Gamma(b)^2}\int_0^1\int_0^1 (st)^a(s^\alpha - t^\alpha)(s^\beta - t^\beta)K_m(s,t)\,d\nu_b(s)\,d\nu_b(t) < 0.$$

The integrand apart from its leading minus sign is positive off the diagonal, a set of full product measure. All integrals are finite. At $b = 1$, $g_1(t) = 1$ and $f_1(a;x) = 1/(1-x)$ is independent of a, so all the coefficients vanish. This completes the theorem.

Verification and source status

The proof is analytic and does not depend on numerical sign tests. The accompanying verifier checks the finite-sum differential identity with exact rational arithmetic, the homogeneous-series recurrence, and independent numerical instances of the integral identities. The live MathDB record, published statement and public claim sources were checked on 11 September 2026; no matching full solution claim was located. The dated records and search limits are in `literature-audit.md`.

Tasks 2–5

2. Determine which nonnegative sequences (c_n) preserve the result when $g_b(a+n)$ is replaced by $c_n g_b(a+n)$, including the full class of log-concave sequences without internal zeros.

3. For $b > 1$, derive explicit positive lower bounds and matching upper bounds for each coefficient ψ_m, in terms of the parameters and its order.

4. Prove or refute the analogous unrestricted coefficientwise log-concavity conjecture for shifts in the second parameter of $I_x(a,b)$.

5. Establish a q-gamma analogue of the finite-sum inequality in Lemma 1 and determine the corresponding parameter ranges.

References

[1] D. B. Karp, *Normalized incomplete beta function: log-concavity in parameters and other properties*, Journal of Mathematical Sciences **217** (2016), 91–107, Conjecture 2, p. 99, and Corollary 1, p. 101. doi:10.1007/s10958-016-2958-z. Published full text; arXiv:1509.05120.